

## Expert Participation Consent Form

**Study Title:** Evaluation of Pharmaceutical Reverse Logistics Alternatives Using a Two-Stage Pythagorean Fuzzy MCDM Framework

**Principal Investigator:** Tolga Kudret KARACA, Ph.D. İstanbul Topkapı University, Department of Industrial Engineering

This study aims to evaluate pharmaceutical reverse logistics alternatives (third-party take-back, manufacturer-led returns, pharmacy-led collection, and government recycling centers) based on 18 criteria grouped under six clusters: economic, environmental, regulatory, operational, accessibility, and infrastructural dimensions. Within the proposed two-stage Pythagorean fuzzy multi-criteria decision-making framework, the Indifference Threshold-based Attribute Ratio Analysis (ITARA) is used to determine the criteria weights, and the Evaluation based on Distance from Average Solution (EDAS) method is applied to rank the alternatives. Sensitivity analysis is performed under six different weighting scenarios and one equal-weight scenario.

### What Participation Involves:

As an expert participant, you are invited to evaluate and rate pharmaceutical reverse logistics alternatives based on your professional experience.

- You will be asked to read a short description of the criteria and alternatives.
- You will then complete the expert evaluation form by indicating the relative importance (weights) and/or membership degrees for each criterion based on the Pythagorean fuzzy set format.
- You may also evaluate the alternatives if requested.
- Participation is expected to take approximately **15–25** minutes, and the session can be completed in one sitting.
- All data collected will be anonymized; no personal identifiers will appear in reports or publications.
- For transparency, anonymized and aggregated results may be shared for scientific or archival purposes.

Your participation is voluntary. You may withdraw from the study at any time without providing a reason. Withdrawal will not affect any current or future professional relationships.

Findings from this study may be published in academic journals, conferences, or policy briefs in an anonymized format. Please indicate your consent preferences below:

- ☐ I consent to the scientific reuse of my anonymized data.
- ☐ I consent to the archival storage of my anonymized data in open data repositories.
- ☐ I consent to the use of my data only for this specific study.

If you have any questions, concerns, or complaints about this research, please contact the principal investigator at **[tolgakudretkaraca@topkapi.edu.tr](mailto:tolgakudretkaraca@topkapi.edu.tr)**

For ethical concerns, you may also contact **[onuroman@topkapi.edu.tr](mailto:onuroman@topkapi.edu.tr)**

**Participant Statement:**

I have read and understood the above information regarding this study. I voluntarily agree to participate as an expert in this research. I understand that my responses will remain confidential and used only for academic purposes.

**Participant Name:** \_\_\_\_\_**Signature:** \_\_\_\_\_**Date:** \_\_\_\_\_**Researcher Name:** \_\_\_\_\_**Signature:** \_\_\_\_\_**Date:** \_\_\_\_\_